

TABLE I. Original Time Observations

Temp. 22.6°C			Temp. 50°C			Temp. 100°C		
Test No.	Press. lb./sq.in.	To sec.	Test No.	Press. lb./sq.in.	To sec.	Test No.	Press. lb./sq.in.	To sec.
1	0	6.13	11	0	4.60	20	0	3.60
2	12000	20.5	12	3000	5.17	21	3000	4.00
3	0	6.27	13	6000	5.57	22	6000	4.17
4	12000	20.7	14	9000	6.87	23	9000	4.70
5	0	6.50	15	12000	8.20	24	12000	4.87
6	3000	8.13	16	9000	7.57	25	9000	4.47
7	6000	11.0	17	6000	6.10	26	6000	4.20
8	9000	15.5	18	3000	5.20	27	3000	4.00
9	12000	21.3	19	0	4.60	28	0	3.67
10	0	6.60						

Corrections for acceleration were made by equation (1) (Reference 1) and were never as large as 1 per cent. A re-examination of the derivation of this formula showed that the constant k is independent of the ball diameter, directly proportional to the sine of the angle above the horizontal, and inversely proportional to the length of path through which the ball moves. The same value as before was therefore used. Corrections for change of path due to compression of the rubber washers were also made as before, and the combined correction for acceleration and compression was never greater than 1 per cent.

From the corrected time values and the density values ac-